(01-15-26)

IT Project Management. [Class One]

Prof. Williamson

— Assignments open Fridays

Module One Should open (01-16-26).

Chapter One: Intro to IT Project Management

Advantages of Using Formal Project Management:

- Better control of financial, physical, and human resources.
- Improved customer relations.
- Shorter development times.
- Lower costs + improved reliability.
- Higher profit margins.
- Better internal Coordination.
- Positive impact on meeting strategic goals
- Higher Worker Morale.

What is a project?

- A temporary endeavor undertaken to create a unique product, service, or result.
- Operations is work done to sustain the business.
- Projects end when their objectives have been reached or the project has been terminated.

Project Attributes

- A Project
  - ↳ Has a unique purpose
  - Temporary
  - Is everchanging in drives
  - Developed using progressive elaboration
  - Requires resources from various areas
  - Should have primary customer or sponsor
    - ↳ The sponsor provides direction and funding

- Project Managers work with project sponsors, team, and other people involved in a project to achieve project goals.

- Project Management is the application of knowledge, skills, tools, and techniques to project activities to meet project requirements.

- Project managers strive to meet the triple constraint (project scope, time, and cost goals) and also facilitate the entire process to meet the needs and expectations of stakeholders.

- Stakeholders are the people involved in or affected by project activities. This includes: the sponsor, banks and other financial institutions, the project manager, the project team, suppliers, opponents to the project.

Project Management Knowledge Areas

Knowledge areas describe the key competencies that project managers must develop.

Project Managers must have knowledge and skills in all ten knowledge areas (scope, schedule, cost, quality, resource, communication, risk, procurement, stakeholder, project integration management.

Project Management tools and techniques

↳ These assist project managers and their teams in various aspects of project management.

Project Success

↳ There are several ways to define project success:

- Project met scope, time, and cost goals.
- The project satisfied the customer/sponsor.
- The results of the project met its main objective such as making or saving a certain amount of money, providing a good return on investment, or making sponsors happy.

Top three reasons why Federal technology projects succeed.
- Adequate funding
- Staff expertise
- engagement from all stakeholders

Research findings show that companies that excel in project delivery capability
- use an integrated toolbox
- grow project leaders
- develop a streamlined project delivery process
- Measure project health using metrics, like customer satisfaction or return on investment

Two important concepts that help projects meet enterprise goals:
- use of Programs
- project portfolio management

A program is "a group of related projects managed in a coordinated manner to obtain benefits and control not available from managing them individually.

↳ Common examples in the IT field are: infrastructure, application development, user support.

A program manager provides leadership and direction for the project managers heading the projects within the program

As part of project portfolio management, organizations group and manage projects and programs as a portfolio of investments that contribute to the entire enterprise's success.

Portfolio managers help their organizations make wise investment decisions by helping to select and analyze projects from a strategic perspective

A best practice is "an optimal way recognized by industry to achieve a stated goal or objective.

Organizations should follow basic principles of project management, including:
- Making sure projects are driven by strategy.
- Engage your stakeholders at all times of a project

Role of the Project Manager

Project Managers must work closely with the other stakeholders on a project, especially the sponsor and project team.

↳ Job descriptions vary but often include responsibilities such as planning, scheduling, coordinating, and working with people to achieve project goals

Project Management is a skill needed in every major IT field from database administrator to network specialist to technical writer

Suggested Skills for Project Managers

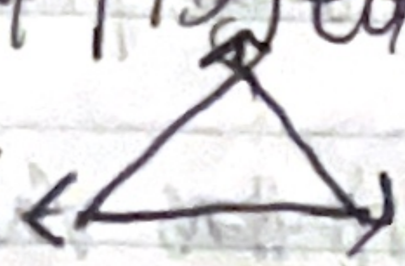
- Project Management body of Knowledge.
- Application area Knowledge, standards, and regulations.
- Project environment knowledge.
- General Management knowledge and skills.
- Soft skills or human relations skills.

Six traits of highly effective project managers are:

- Be strategic business partner
- Encourage and recognize valuable contributions
- Respect and motivate stakeholders
- Be fully vested in success
- Stress integrity and accountability
- Work in or through ambiguity.

The talent triangle includes:

Technical Project Management Skills



Leadership Skills

Strategic and business management skills.

Leadership Styles:

- Laissez-faire
- Transactional
- Servant Leader
- Transformational
- Charismatic
- Interactional

Careers for IT project Managers

1. Full Stack Software Development
2. Project Management
3. Cyber Security
4. Networking
5. User Experience / User Interface (UX/UI) Design
6. Quality Assurance (QA) / Testing
7. Cloud Engineering
8. Big Data
9. Machine Learning / Artificial Intelligence
10. DevOps

A **Project Management Office (PMO)** is an organizational group responsible for coordinating the project management function throughout an organization

Ethics in Project Management

Ethics, loosely defined, is a set of principles that guide our decision making based on personal values of what is "right" and "wrong".

Project Management Software

↳ Three Main Categories:

- Low-End Tools: Handle single or small projects well, cost under $200 per user.
- Mid-Range Tools: Handle multiple projects and users, cost $200 - $1,000 per user. Microsoft Project is the most popular.
- High-end Tools: AKA enterprise project management software, often licensed on a per-user basis.

Class Cancelled Next Week. 1-29-26

Chapter 2: Project Management and Information Technology Context

Systems view of proj. man.

Projects must operate in a broad organizational environment. Project managers need to use systems thinking:
↳ Taking a holistic view of carrying out projects within the context of the organization.

What is a systems Approach?

A systems approach emerged in the 1950s to describe a holistic and analytical approach to management and problem solving.

Three parts include:

- Systems Philosophy: an overall model for thinking about things as systems.
- Systems Analysis: problem solving approach.
- Systems Management: Address business, technological, and organizational issues before making changes to systems.

Understanding Organizations

Systems approach requires that project managers always view their projects in the context of the larger organization.

Organizational issues are often the most difficult part of working on and managing projects.

To improve the success rate of IT projects it is important to understand people as well as the organization.

The Four Frames of Organizations:

- Structural Frame → Roles, responsibilities, coordination, control.
- Human Resource Frame → providing harmony between needs of the organization and needs of the people.
- Political Frame → Coalitions composed of varied individuals and interest groups
- Symbolic Frame → Symbols and meanings related to events

Culture within organizations is often to blame for project failures

Organizational Structures
- Functional → Managers report to CEO
- Project → Program Managers report to CEO
- Matrix → personnel report to two or more bosses; can be a weak/balanced/strong Matrix.

Organizational Culture
Culture is a set of shared assumptions, values, and behaviors that characterize the functioning of an organization.

↳ Ten Characteristics of Organizational Culture:

- Member Identity
- Group Empathy
- people Focus
- Reward Criteria
- Means-End Orientation
- Unit Integration
- Control
- Risk Tolerance
- Conflict Tolerance
- Open-systems Focus

Focusing on Stakeholder Needs

- Project Managers must identify, understand, and manage relationships with all stakeholders
- Using the Four Frames of organization can help meet needs and expectations.

Importance of Top Management Commitment
people in top management positions are Key Stakeholders in projects

A very important factor in successful projects is the level of commitment and support received from top management. Many projects will fail without top management commitment.
↳ A champion is a senior manager who acts as a key proponent for a project.

IT Governance addresses the authority and control for key IT activities in organizations, including IT infrastructure, IT use, and project management. A lack of IT governance can be dangerous.

The Need for Organizational Commitment to Information Technology.

- If the organization has a negative attitude toward IT, it will be difficult for an IT project to succeed.
- Having a Chief Information Officer (CIO) at a high level in the organization helps IT projects
- Assigning non-IT people to IT projects encourages more commitment.

The Need for Organizational Standards

Standards and guidelines help project managers be more effective.

Project/product Life Cycles

It is good practice to divide projects into several phases, as well as developing projects.

A life cycle is a collection of phases that defines:

- what work will be performed in each phase.
- What will be produced and when.
- Who is involved in each phase.
- How management will control and approve work

A deliverable is a product or service produced or provided as part of a project.

A project should successfully pass through each phase in order to continue to the next.

The Context of IT projects

Project context has a critical impact on which product development lifecycle will be most effective for a particular project.

IT projects can be very diverse. Software even more so than hardware.

Agile

↳ An approach where requirements and solutions evolve through collaboration.

Scrum is the leading agile development method for projects with a complex, innovative scope of work